

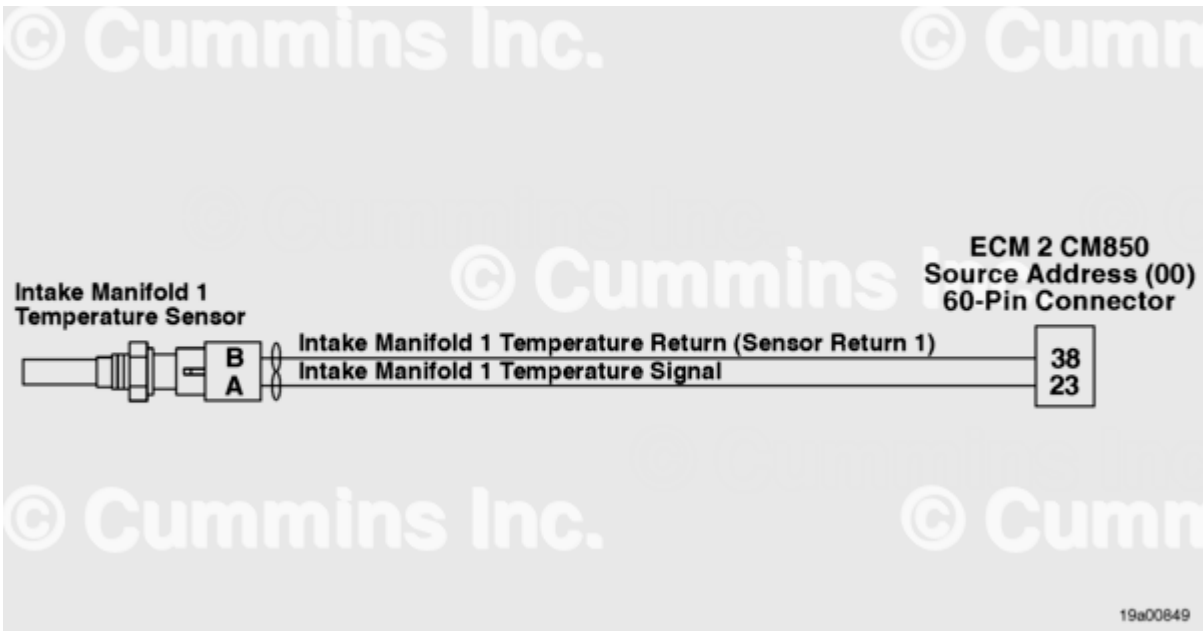
Fault Code: 488

Intake Manifold 1 Temperature - Data Valid But Above Normal Operating Range - Moderately Severe Level

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Overview

Codes	Reason	Effect
Fault Code: 488 PID(P): P105 SPN: 105 FMI: 0/16 Lamp: Amber SRT:	Intake manifold 1 temperature above engine protection warning limit.	Possible reduced engine performance.



Intake Manifold 1 Temperature Sensor Circuit

Circuit Description

The intake manifold temperature sensor is a variable resistor sensor used by the ECM to monitor the intake manifold temperature. The intake manifold temperature sensor has two circuits: signal, and return circuits. The signal voltage indicates the intake manifold temperature.

Component Location

The intake manifold 1 temperature sensor is located in the left bank rear air intake manifold.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the generator set controller is active or when the engine is running.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected the intake manifold 1 air temperature sensor was greater than the engine protection limit.

Action Taken When The Fault Code Is Active

- The generator set controller displays a warning fault immediately when the diagnostics runs and fails.
- The torque output of the engine will be reduced.

Conditions For Clearing The Fault Code

- To validate the repair, bring the engine up to operating temperature and run it in normal loaded conditions for 15 minutes.
- The generator set controller will turn off the warning indicator immediately after the user presses reset.
- The "Reset All Faults" command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

There are multiple ECMs for the engine models included in this manual. Each ECM has an individual source address that displays when Cummins® electronic service tool or equivalent is connected. When troubleshooting a fault code, use the source address displayed in Cummins® electronic service tool or equivalent to determine which ECM and circuit is affected.

Possible causes of this fault code include:

- Restricted air flow through the charge-air cooler
- Low coolant level
- High coolant temperature
- Undersized aftercooler
- High turbocharger compressor outlet temperature.

Refer to Troubleshooting Fault Code 488 . (</qs3/pubsys2/xml/en/procedures/60/60-t05-488.html>)

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